Dividing Numbers in Scientific Notation

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Scientific Notation

Scientific notation expresses a number in the form

$$a \times 10^n$$

where $1 \leq a < 10$ and n is an integer.

Dividing Formula

To divide numbers in scientific notation, use this formula:

$$\frac{a \times 10^m}{b \times 10^n} = \frac{a}{b} \times 10^{m-n}$$

Remember: Divide, Subtract, Normalize!

Divide the coefficients.

Subtract the exponents.

Normalize the result.

Step 1: Divide the Coefficients

Divide the first coefficient by the second coefficient.

$$\frac{8 \times 10^7}{2 \times 10^3}$$

$$\frac{8}{2} = 4$$

Step 2: Subtract the Exponents

Subtract the exponent of the denominator from the exponent of the numerator.

$$\frac{8 \times 10^7}{2 \times 10^3}$$

$$10^{7-3} = 10^4$$

Top Minus Bottom

Part	**Operation**
Coefficients	$\frac{\text{top}}{\text{bottom}}$
Exponents	$\text{top} - \text{bottom}$

Step 3: Simplify

Express the result in proper scientific notation, if necessary.

The coefficient must be at least 1 but less than 10.

$$1 \leq a < 10$$

Adjusting the Coefficient

When the coefficient is not in proper scientific notation, adjust the coefficient and the exponent together.

$$0.5 \times 10^3 = 5 \times 10^2$$

Examples

Example 1

Simplify.

$$\frac{8 \times 10^7}{2 \times 10^3}$$

$$= \frac{8}{2} \times 10^{7-3}$$

$$= 4 \times 10^4$$

Example 2

Simplify.

$$\frac{9.6 \times 10^6}{3 \times 10^2}$$

$$= \frac{9.6}{3} \times 10^{6-2}$$

$$= 3.2 \times 10^4$$

Example 3

Simplify.

$$\frac{2 \times 10^5}{4 \times 10^2}$$

Divide and subtract:

$$= 0.5 \times 10^3$$

Example 3 (continued)

Normalize:

$$0.5 \times 10^3 = 5 \times 10^{-1} \times 10^3$$

$$= 5 \times 10^2$$

Example 4

Simplify.

$$\frac{7.2 \times 10^{-4}}{1.8 \times 10^{-7}}$$

$$= \frac{7.2}{1.8} \times 10^{-4-(-7)}$$

$$= 4 \times 10^{3}$$

Example 5

Simplify.

$$\frac{3 \times 10^2}{6 \times 10^5}$$

Divide and subtract:

$$= 0.5 \times 10^{-3}$$

Example 5 (continued)

Normalize:

$$0.5 \times 10^{-3} = 5 \times 10^{-1} \times 10^{-3}$$

$$= 5 \times 10^{-4}$$

Practice Exercises

Practice 1

Divide the following numbers expressed in scientific notation.

$$\frac{6 \times 10^8}{2 \times 10^3}$$

Practice 2

Divide the following numbers expressed in scientific notation.

$$\frac{8 \times 10^9}{4 \times 10^5}$$

Practice 3

Divide the following numbers expressed in scientific notation.

$$\frac{9 \times 10^7}{3 \times 10^2}$$

Practice 4

Divide the following numbers expressed in scientific notation.

$$\frac{4.8 \times 10^6}{1.2 \times 10^2}$$

Practice 5

Divide the following numbers expressed in scientific notation.

$$\frac{7.5 \times 10^8}{2.5 \times 10^4}$$

Practice 6

Divide the following numbers expressed in scientific notation.

$$\frac{3.6 \times 10^5}{6 \times 10^2}$$

Practice 7

Divide the following numbers expressed in scientific notation.

$$\frac{8.4 \times 10^9}{2.1 \times 10^3}$$

Practice 8

Divide the following numbers expressed in scientific notation.

$$\frac{5 \times 10^4}{2 \times 10^7}$$

Practice 9

Divide the following numbers expressed in scientific notation.

$$\frac{6.4 \times 10^{-3}}{1.6 \times 10^{-6}}$$

Practice 10

Divide the following numbers expressed in scientific notation.

$$\frac{9.6 \times 10^{-4}}{3.2 \times 10^{2}}$$

Activities

Activity 1

Divide the following numbers expressed in scientific notation.

$$\frac{8 \times 10^9}{4 \times 10^4}$$

Activity 2

Divide the following numbers expressed in scientific notation.

$$\frac{6 \times 10^8}{3 \times 10^5}$$

Activity 3

Divide the following numbers expressed in scientific notation.

$$\frac{8 \times 10^7}{2 \times 10^2}$$

Activity 4

Divide the following numbers expressed in scientific notation.

$$\frac{6.4 \times 10^7}{1.6 \times 10^3}$$

Activity 5

Divide the following numbers expressed in scientific notation.

$$\frac{9 \times 10^8}{3 \times 10^4}$$

Activity 6

Divide the following numbers expressed in scientific notation.

$$\frac{4.8 \times 10^6}{8 \times 10^2}$$

Activity 7

Divide the following numbers expressed in scientific notation.

$$\frac{7.2 \times 10^9}{2.4 \times 10^3}$$

Activity 8

Divide the following numbers expressed in scientific notation.

$$\frac{6 \times 10^5}{3 \times 10^8}$$

Activity 9

Divide the following numbers expressed in scientific notation.

$$\frac{8.1 \times 10^{-2}}{2.7 \times 10^{-5}}$$

Activity 10

Divide the following numbers expressed in scientific notation.

$$\frac{8.4 \times 10^{-3}}{2.8 \times 10^{2}}$$

Thank you!